

Summary

Software engineer and data scientist with experience shipping production systems at Marriott and published work in scientific computing. Pursuing an M.S.E. at Penn to deepen expertise at the intersection of data science, ML, and scalable software.

Education

University of Pennsylvania: 2026-2028 | M.S.E. in **Scientific Computing** |

Virginia Tech: 2023 - 2026 | B.S. in **Computational Modeling and Data Analytics** | GPA: 3.85 | Summa Cum Laude |

Professional Experience

Marriott

Associate Software Engineer | 08/2024 - 06/2026

- Lead developer for the provisioning automation service project for internal company wide use in Java SpringBoot. Lowered provisioning times from days to minutes for 250+ APIs and client access to APIs and developed a UI for internal use.
- Developed Python based MCP servers and mock Marriott APIs for internal POC of agentic API capabilities allowing for booking of hotel rooms, hotel room detail queries, and caching of request-responses for quick LLM-bypassing responses through n8n.
- Hosted internal meetings to verify requirements, procure feature requests, and get feedback on current functionality which led to new production API and client data viewing service capabilities.
- 50+ users are currently utilizing the service to automate provisioning of clients, APIs, and the viewing of this data across environments. This has led to saving 20-30 hours per month across a variety of internal teams.
- Collaborated with DevOps to deploy the microservice as well as revamped user interface through Harness, allowing all internal users to access and utilize the provisioning service to enable company wide time saving.

Core Platform Software Engineering Intern | 06/2024 - 08/2024

- Built a SpringBoot microservice to automate API provisioning across development lifecycle environments and client channels.
- Reduced the current processes from 30+ manual actions for the provisioning of API's to 4 actions.
- Developed a POC User Interface using React to demo the automation microservice to stakeholders and demoed this to leadership.

Virginia Tech

Machine Learning Dimensionality Reduction Undergraduate Researcher | 08/2024 - 06/2026

- Developed autoencoder based methods for ODE input data to create a lower dimensionality representation for inverse problem mapping focused neural networks, reducing the computational cost of training the inference neural networks.
- Achieved a 99%+ accuracy on parameter predictions with autoencoder lower dimensionality data.

Inverse Problem Machine Learning Solutions Undergraduate Researcher | 02/2025 - 06/2026

- Research into solving physical modeling inverse problems for the FitzHugh-Nagumo ODE and the Hodgkins Huxley model using different neural network architectures in PyTorch, allowing for efficient multi-parameter prediction with a 97%+ accuracy.

Code Generation Toolkit Undergraduate Researcher | 09/2023 - 07/2024

- Utilized tree graph theory methodology to create a code generation platform using Python to reduce algorithmic variants in scientific computing, named *CodeGenerationKit* (CGKit), in collaboration with Argonne National Laboratory.
- Code generation platform is utilized in the *Flash-X* computing platform, leading to a 55%+ reduction in overall algorithm code, improving readability and development efficiency. This work was published in *Future Generation Computer Systems*.

Technical University of Munich

Computational Geophysics Researcher | 07/2025 - 08/2025

- Developed new coupled Tsunami Landslide simulations using Peano4 using Python and C++ utilizing finite volume and ADER-DG PDE solvers. Presented this work to TUM leadership, paving the way for real world tsunami-landslide comparisons.
- Developed limited solvers allowing for a combination of finite volume and ADER-DG solutions to reduce computational cost.

Skills

Languages: Python, Java, SQL, R, C

Technologies: Spring, Harness, Git, Numpy, Pandas, Scikit